

Name of Student: _____

Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

End-Term Examination Nov - 2025

dr. yatendra sahu

Course : B.Tech

Semester:- 3rd

Branch: CSE/AI/CPS/DS/CS

Subject Code : CSE/AI/CPS/DS/CS - 3002

Max. Marks:60

Subject Name: Database Management System

Time : 03hr

NOTE:

1. All the questions are compulsory. 2. While solving write question number and its part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.

- 1.A. i. Draw the component modules of a DBMS and explain in detail how they interact with each other during a database operation such as executing a query or performing a transaction. (6)
- ii. A database has 100,000 records (100 bytes each), block size = 1,000 bytes (10 records per block). Compare Ordered Indexing (B+ Tree) and Hashing techniques in terms of search cost, insertion, and range queries, using suitable numerical assumptions.

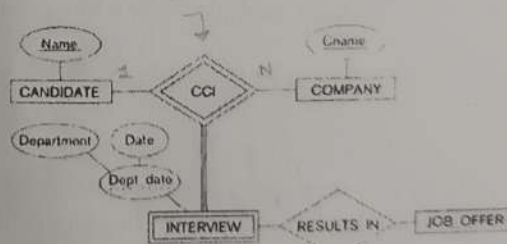
B. Let E and W denote entity types in an Entity-Relationship (ER) schema. (6)

- i. Specify the conditions under which W is considered a weak entity type in terms of the nonexistence of a unique key within its own attribute set.
- ii. Provide formal definitions (using symbols and set notation where appropriate) for:
 - the owner entity type E ,
 - the weak entity type W ,
 - the identifying relationship type $R \subseteq E \times W$, and
 - the partial key $P_W \subseteq \text{Attr}(W)$.

- 2.A. i. Prove that relational algebra (RA) and tuple relational calculus (TRC) are equally expressive, i.e., for every RA expression E , there exists a TRC formula F such that both define the same relation, and vice versa. (6)

- ii. A database is being constructed to keep track of the teams and games of a sports league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game, and the result of the game. Design an ER schema diagram for this application, stating any assumptions you make.

B. Consider the ER diagram given in figure to convert it into relational tables. (6)



- 3.A. Consider the given schema to write sql queries for below given questions: Sailors(sid, sname, rating, age), Boats(bid, bname, color), Reserves(sid, bid, rdate) (6)

- i. Find the colours of boats reserved by the sailor named 'Albert'.
- ii. Find all sailor IDs of sailors who have a rating of at least 8 or who have reserved boat 103.
- iii. Find the names of sailors who have *not* reserved a boat whose name contains the string 'storm'.
- iv. Find the average age of sailors for each rating level.
- v. Find the sailor(s) who have made the maximum number of reservations.
- vi. Find sailors whose rating is higher than the average rating of sailors who reserved a red boat.

B. Consider the given schema to write relation algebra for below given questions: Sailors(sid, sname, rating, age), Boats(bid, bname, color), Reserves(sid, bid, rdate) (6)

- Find all sailor IDs of sailors who have a rating ≥ 8 or have reserved boat 103.
- Find the sailor IDs of sailors older than 20 who have not reserved a boat whose name includes "thunder".
- Find the names and ages of sailors who have reserved red boats.
- Find the sailor IDs of sailors who have reserved red boats but not green boats.
- Find the sailor IDs of sailors who have reserved red boats but not green boats.
- Find the names of sailors who have reserved at least one boat of every color available in the Boats relation.

4.A. i. Consider two sets of functional dependency. (6)

$$F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\} \quad E = \{A \rightarrow CD, E \rightarrow AH\}.$$

Clearly show all intermediate steps to check are they Equivalent?

ii. Consider a relation R(A, B, C, D, E, F) with the following set of functional dependencies:

$$E = \{AB \rightarrow D, B \rightarrow A, D \rightarrow A, A \rightarrow C, C \rightarrow F, CF \rightarrow E\}.$$

Construct the canonical cover m for the set of functional dependencies E.

B. A relation R(P, Q, R, S, T, U, V, W) satisfies the following set of Functional Dependencies (FDs): (6)

$$F = \{P \rightarrow QR, R \rightarrow S, PU \rightarrow V, PR \rightarrow T, Q \rightarrow W, V \rightarrow U\}.$$

The relation R is decomposed into the following four sub-relations:

$$R_1(P, Q, R, T), R_2(R, S), R_3(U, V, P), R_4(Q, W)$$

- Determine whether the decomposition is lossless or lossy.
- Check whether the decomposition is dependency preserving.

5.A. Consider a relation R(A, B, C, D, E, F) with the following set of functional dependencies (FDs): (6)

$$F = \{AB \rightarrow C, AD \rightarrow E, B \rightarrow D, E \rightarrow A, C \rightarrow F, F \rightarrow B\}$$

- Find all candidate keys of relation R.
- Determine the highest normal form that R satisfies (1NF, 2NF, 3NF, or BCNF)
- If the relation is not in BCNF, decompose it step-by-step into BCNF relations.

B. Consider the following schedule S involving five transactions (T_1 – T_5) and data items X, Y, Z, P, Q, R: (6)

$$S: r_1(X), w_3(X), r_2(Y), w_1(Y), r_4(Y), r_3(Z), w_5(Y), r_5(X), w_2(P), r_1(P), w_4(X), r_3(P), w_5(Z)$$

- Construct the precedence graph and determine whether the schedule is conflict serializable.
- Check whether the schedule is view serializable if it is not conflict serializable.

1 (A) i)

5 (6)